

DESIGN AND IMPLEMENTATION OF A DATABASE MANAGEMENT SYSTEM FOR AIRLINE RESERVATION AND FLIGHT OPERATIONS

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PROBLEMS

MANUAL RECORD MANAGEMENT

- Maintaining passenger and flight records manually is time-consuming and inefficient.
- Updating booking, passenger, and flight information can be difficult to track.
- Manual records increase the risk of lost or outdated information.

DUPLICATE BOOKING AND SEAT CONFLICTS

- Multiple passengers may accidentally be assigned the same seat.
- Incorrect booking records can create seat allocation conflicts.
- Duplicate bookings can lead to passenger dissatisfaction and operational issues.

BAGGAGE TRACKING PROBLEMS

- Difficulty in tracking passenger baggage from check-in to arrival.
- Lack of proper tracking can result in lost or misplaced baggage.
- Poor baggage tracking can lead to delays, complaints, and passenger dissatisfaction

SCOPE & OBJECTIVE

PROJECT SCOPE

- Develop a centralized database for airline reservation and flight operations.
- Manage passenger and customer information efficiently.
- Store and manage flight schedules, routes, airports, and aircraft details.
- Manage fares, payments, cancellations, and refunds.
- Maintain data integrity and consistency using database constraints.
- Manage seat availability and seat allocation

OBJECTIVES

- To ensure data accuracy, consistency, and integrity using database constraints.
- To manage payments, cancellations, and refunds efficiently.
- To design a database that is scalable, secure, and easy to maintain.
- To efficiently manage passenger, flight, aircraft, and airport information.
- To develop a centralized database for airline reservation and flight operations.

MAIN ENTITIES

PASSESNGER



Searches flights, makes bookings, selects seats, makes payments, and checks in.

RESERVATION STAFF



Manages bookings, passenger details, tickets, and seat allocation.

FLIGHT OPERATIONS STAFF



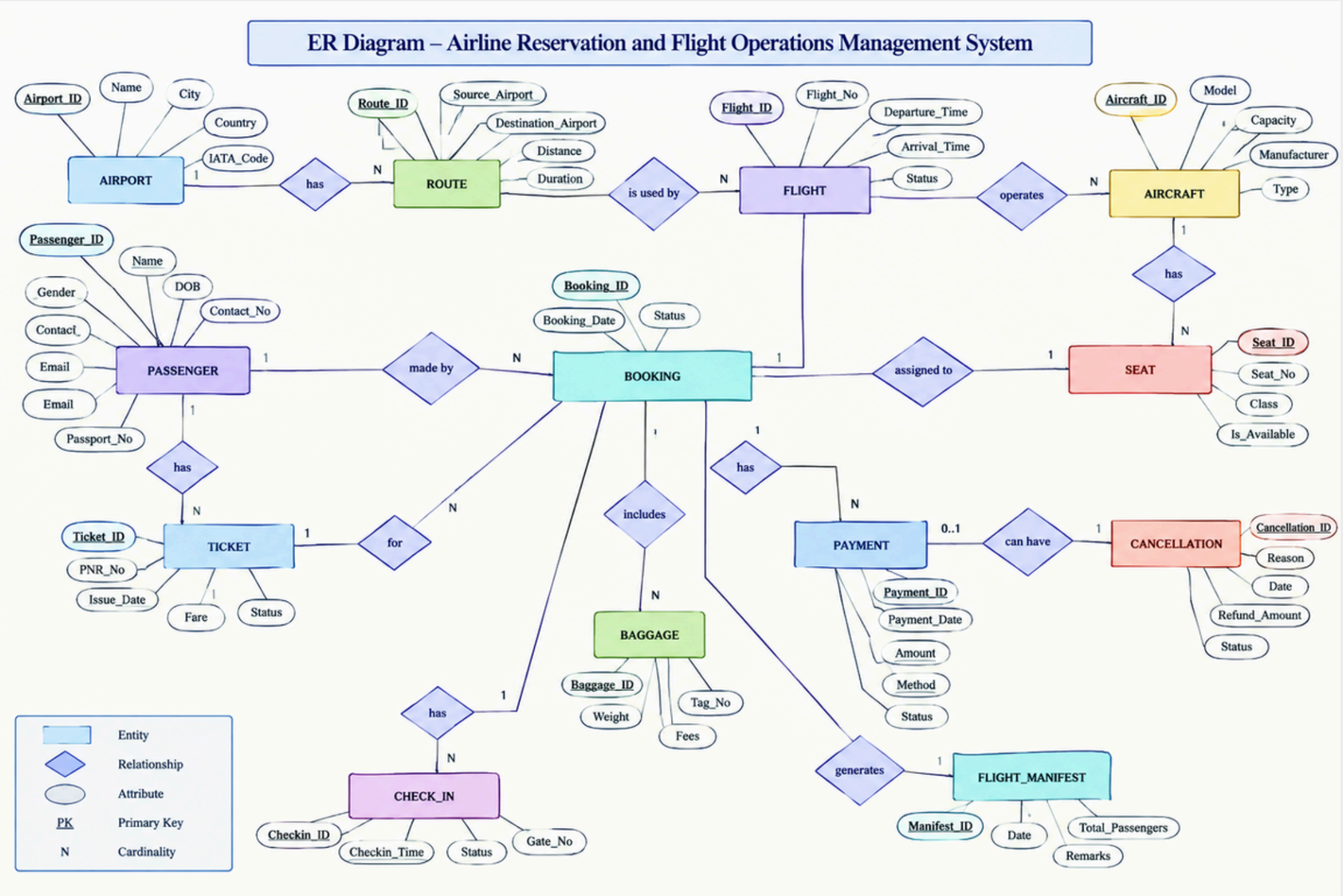
Manages flights, aircraft, routes, schedules, and flight status.

ADMINISTRATOR



Manages the database, users, access permissions, and system records.

ER DIAGRAM



1.Shows the structure of the Airline Reservation and Flight Operations Identifies the main entities such as Passenger, Flight, Aircraft, Airport, Booking, Ticket, Seat, Baggage, and Payment.

2.Shows the relationships between different entities in the system.

3.Shows the connection between flights, routes, airports, and aircraft.

4.Displays primary keys (PK) and foreign keys (FK) used to connect tables.

5.Shows cardinality such as one-to-one and one-to-many relationships.

6.Provides a clear blueprint for converting the system into a relational database.

7.Helps maintain data integrity and reduce data redundancy.

CONCLUSION

- Provides a systematic and efficient airline reservation management system.
- Reduces manual record-keeping and human errors.
- Prevents duplicate bookings and seat conflicts.
- Improves passenger and baggage tracking.
- Enables better coordination between passengers, staff, and administrators.
- Ensures secure and organized storage of reservation data.
- Makes booking, cancellation, and flight management faster and easier.
- Improves overall customer experience and operational efficiency.

THANK YOU

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